Tri-HB Handbook V4

Six loading modes, Streamlit workflow, and damage-validation equations

Beamer presentation generated from Handbook V4

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Presentation Overview

1. Purpose and Apparatus
2. Mechanics Backbone
3. Six Loading Modes
4. Integrated Streamlit Workflow
5. Handbook V4 Deliverable

Purpose and Apparatus

Why Handbook V4 matters

Core purpose

Handbook V4 connects the physical Tri-HB facility, the six loading modes, and the Streamlit digital twin into one reproducible workflow.

Experimental need

Dynamic rock failure is rarely uniaxial. In-situ stress, blast waves, seismic pulses, and excavation disturbance are multiaxial and path dependent.

Digital-twin need

Every calculation must be traceable from setup, through wave reduction, to stress path, damage, energy density, and validation descriptors.

Monash Tri-HB system: one platform, six modes

- X-axis gas-gun train for classical SHPB and coupled triaxial loading.
- Independent Y and Z bar pairs for true triaxial static pre-stress.
- Electromagnetic pulse generators for programmable amplitude, duration, symmetry, and inter-axis delay.
- Current digital-twin pre-stress range: 0–50 MPa per axis.

Physical interpretation

Modes 1–3 preserve gas-gun heritage; Modes 4–6 add clean pulse control and true multidirectional timing studies.

Mechanics Backbone

The equations reduce everything to four checks

1. Wave propagation

Elastic bar-wave mechanics set the usable signal window.

2. Specimen equilibrium

The three-wave method requires transit-time and force-balance checks.

3. Stress path

$\sigma_1, \sigma_2, \sigma_3$ are converted to p, q, and Lode angle θ.

4. Damage and energy

Failure index, stiffness loss, and energy density form the validation layer.

Calibration policy carried into the Streamlit app

Updated DIF convention

Handbook V4 and the simulator presets now use a literature-reference strain-rate scale of

$$\dot{\varepsilon}_{\mathrm{ref}} = 10^{-5}\ \mathrm{s}^{-1}.$$

Consequence

Worked examples, simulator presets, and handbook equations now follow the same calibration policy rather than mixing handbook and app conventions.

$$\mathrm{DIF} = 1 + C \log_{10}\left(\frac{\dot{\varepsilon}}{10^{-5}}\right), \qquad \mathrm{DIF} \le \mathrm{DIF}_{\mathsf{max}}.$$

Six Loading Modes

The six-mode comparison is the handbook's decision map

Mode	Drive	Boundary state	Best diagnostic
1	Gas-gun X	Unconfined	Baseline three-wave response
2	Gas-gun X	Chamber, $\sigma_2 = \sigma_3$	Axisymmetric p–q path
3	Gas-gun X	Active true triaxial	Confinement-strengthened branch
4	EM X	Independent pre-stress	Half-sine rate control
5	EM X/Y/Z	Stress-path timing	Multi-axis stress history
6	EM $\pm$ axes	Symmetric squeeze	Hydrostatic compaction path

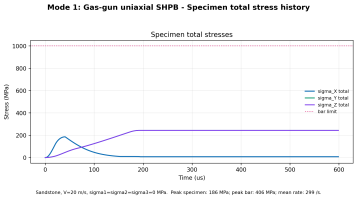
Presentation asset policy

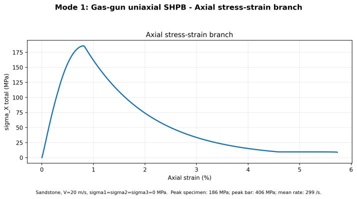
The handbook now exports separate per-mode diagnostics, so slides can use exactly

Mode 1: gas-gun uniaxial SHPB

Use when:

- Baseline dynamic uniaxial strength is the target.
- Published classical SHPB comparison is needed.
- Confinement and Lode-angle effects are outside scope.





Main limitation

Single loading axis, no active confinement.

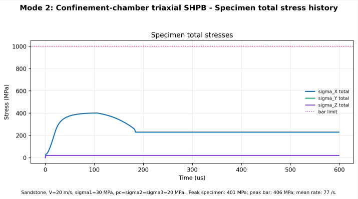
Mode 2: confinement-chamber SHPB

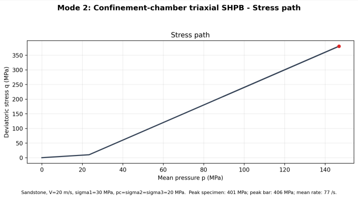
Use when:

- Axisymmetric triaxial confinement is sufficient.
- The specimen is cylindrical or jacketed.
- The stress path can remain at $\theta = 0$.

Main limitation

Lateral pressure is passive; independent σ_2 and σ_3 cannot be imposed.





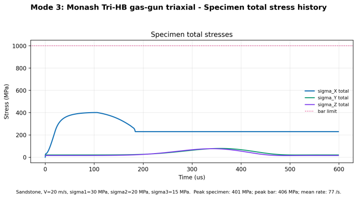
Mode 3: Monash gas-gun true triaxial loading

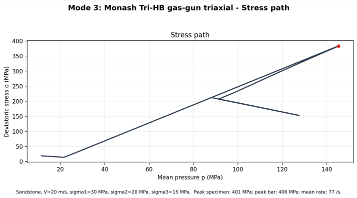
Use when:

- In-situ pre-stress matters.
- σ_2 and σ_3 must be independently controlled.
- Dynamic X loading is combined with true triaxial confinement.

Signature capability

Gas-gun dynamic loading plus active orthogonal bar/cylinder confinement.





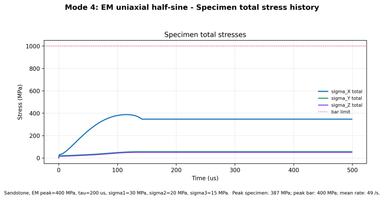
Mode 4: electromagnetic uniaxial impact

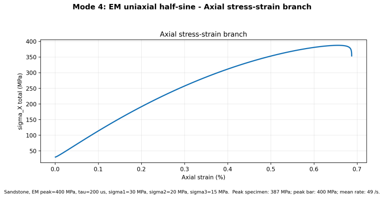
Use when:

- A clean half-sine pulse is preferred.
- Pulse amplitude and duration must be tuned independently.
- Rate effects are the main research question.

Design advantage

The EM system decouples peak stress from pulse duration.





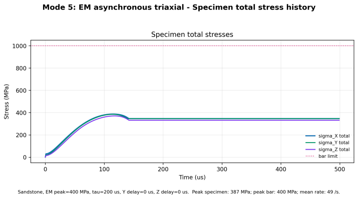
Mode 5: electromagnetic asynchronous triaxial impact

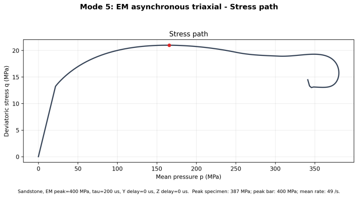
Use when:

- Loading order and inter-axis delay are the physics.
- Crack initiation may depend on which axis arrives first.
- Step 2 delay regime and Step 3 stress path are central outputs.

Interpretation warning

A uniaxial constitutive fit is not meaningful for asynchronous triaxial paths.





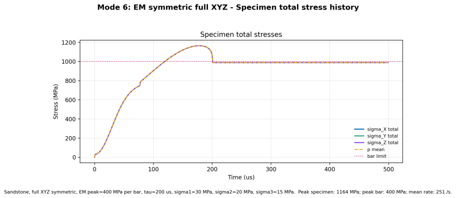
Mode 6: electromagnetic symmetric multidirectional impact

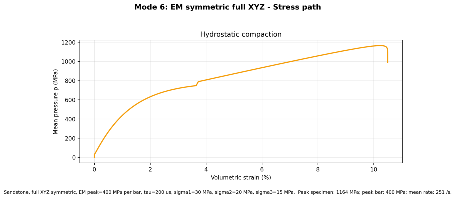
Use when:

- Hydrostatic or near-hydrostatic dynamic loading is required.
- Specimen stress must exceed one-sided bar limits.
- Momentum cancellation is part of the design requirement.

Key idea

Opposing pulses double specimen stress while keeping single-bar stress unchanged.





Mode selection: match the apparatus to the question

Mode	Best question	Primary diagnostic
1	Baseline dynamic strength	Three-wave equilibrium
2	Axisymmetric confinement	Confinement correction and $\theta = 0$
3	In-situ true triaxial pre-stress	Independent $\sigma_1, \sigma_2, \sigma_3$
4	Clean rate sweep	Half-sine amplitude-duration control
5	Loading-order effects	Δt^*, p–q–θ path
6	Hydrostatic cap response	Symmetric energy folding and momentum cancellation

Practical rule

Choose the simplest mode that still preserves the stress state and timing physics required by the research question.

Integrated Streamlit Workflow

The integrated app is the operational front door

Recommended launch

```
py -3.13 -m pip install -r requirements.txt

py -3.13 -m streamlit run tri_hb_integrated.py --server.port 8501
```

On macOS/Linux, replace `py -3.13` with the local launcher, commonly `python` or `python3`.

Loaded modules

- `Tri-HB.py`: Step 1 setup, simulator, export.
- `wave_damage.py`: Steps 2–4 wave, stress path, damage, energy, validation.

Step 1 creates the shared experimental state

Inputs

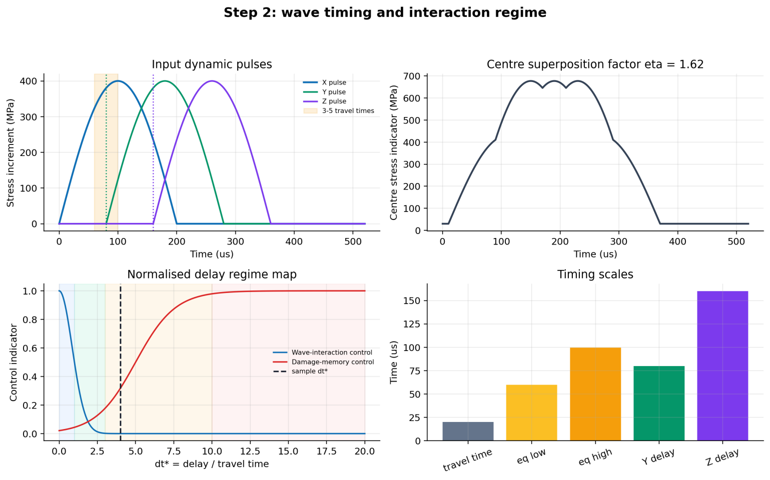
- material preset and specimen geometry
- bar cross-section and elastic properties
- loading mode, pulse source, and pre-stress
- simulator or uploaded gauge data

Session-state handoff

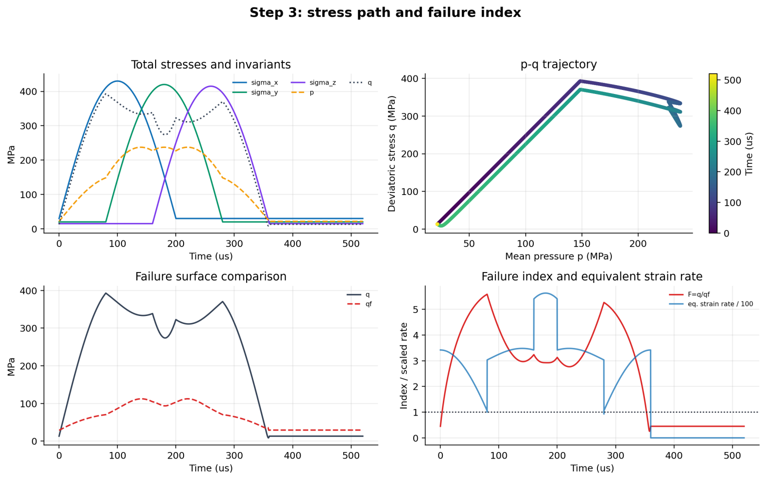
Shared keys
tri_hb_latest_config
tri_hb_latest_result

These keep later steps synchronised with the current setup.

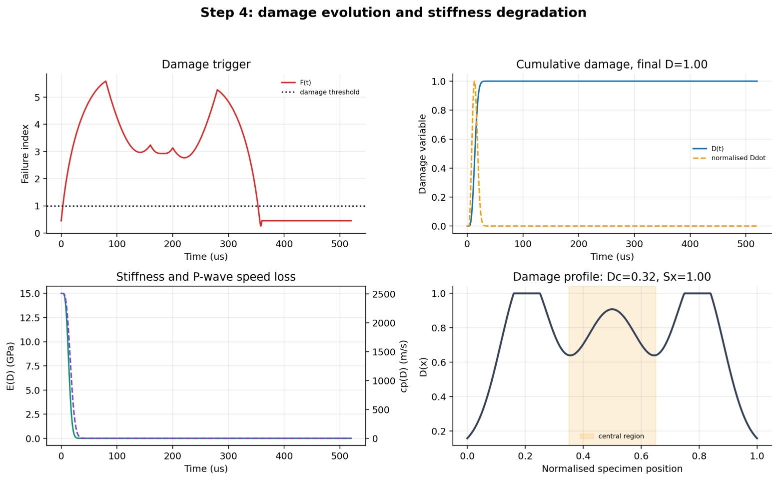
Step 2: wave timing and delay regime



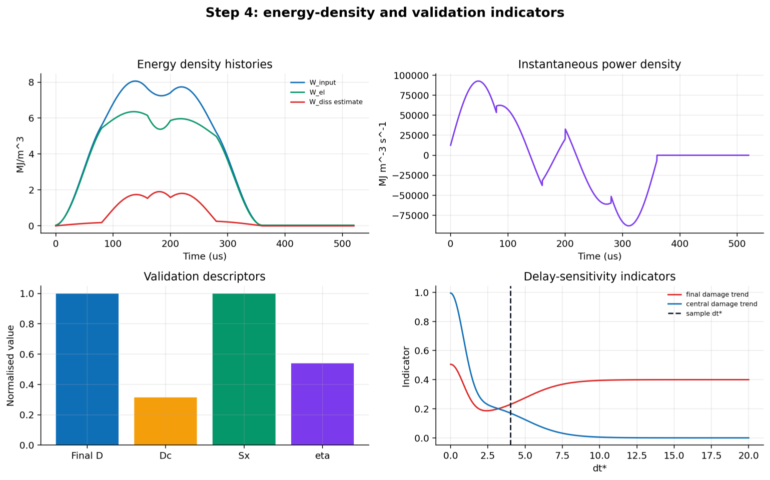
Step 3: stress path and failure index



Step 4: damage evolution and stiffness loss



Step 4: energy density and validation descriptors



Outputs connect the handbook to experiments

Exportable data

- stress–strain histories
- strain-rate histories
- incident/reflected/transmitted energies
- Step 2–4 CSV/XLSX summaries

Validation targets

- bar gauge amplitude and delay
- DIC localisation
- CT damage volume and crack orientation
- DEM bond breakage and dissipated energy

Handbook V4 Deliverable

What is new in this presentation-ready V4

- Simulator presets and handbook equations are aligned around $\dot{\varepsilon}_{\mathrm{ref}} = 10^{-5}\ \mathrm{s}^{-1}$.
- Six-mode figures are inserted into the corresponding apparatus chapters.
- Step 2–4 figures now document wave timing, stress path, damage, energy density, and validation descriptors.
- The launch instructions use the tested Windows launcher: `py -3.13 -m streamlit`.
- Long inline code keys were reformatted to avoid PDF margin overflow.

Recommended use in teaching and research

Teaching sequence

1. start with Mode 1 wave reduction
2. add confinement through Modes 2–3
3. move to EM pulse control in Modes 4–6
4. finish with Step 2–4 damage validation

Research sequence

1. choose mode from the stress-path question
2. check bar plasticity and equilibrium
3. export reduced histories
4. calibrate damage and DEM descriptors

Closing

Handbook V4 is now a linked system.

Apparatus modes, equations, Streamlit workflow, figures, and validation outputs all point to the same calibration story.